Question:

A 34-year-old poultry worker presents to his physician with a sore throat and a non-productive cough for 2 weeks. His cough is associated with fever. The vital signs include: blood pressure 120/80 mm Hg, heart rate 67/min, respiratory rate 18/min, and temperature 37.6°C (98.0°F). Physical examination shows oropharyngeal erythema and scattered, moist rales on lung auscultation. The patient’s X-ray demonstrates patchy reticular opacities in the perihilar regions of both lungs. After some additional tests, he is diagnosed with community-acquired pneumonia and is treated with cephalexin with modest improvement. Which of the following best describes the immune response elicited by the pathogen that is causing this patient’s condition?Answer Choices:A. It activates TLR5 on the surface of macrophages.B. It primarily induces the Th1-cell response.C. Peptidoglycan is its major antigen that induces an immune response.D. This pathogen evades the immune response by encapsulation.

Reasoning + Answer:

<think> The patient's condition is community-acquired pneumonia that was treated with cephalexin with modest improvement. Cephalexin is a bacteriostatic antibiotic that is effective against a variety of gram-positive and gram-negative bacteria. The fact that the patient responded to cephalexin suggests that the causative pathogen is likely a bacterium. Among the options provided, the one that is most relevant to the immune response elicited by a bacterial pathogen is the one that involves peptidoglycan. Peptidoglycan, also known as murein, is a component of the bacterial cell wall and is recognized by the immune system as a pathogen-associated molecular pattern (PAMP). It activates the immune response by inducing the production of inflammatory cytokines and the activation of macrophages. This is most closely related to option C, which states that peptidoglycan is its major antigen that induces an immune response. Options A and B are related to the immune response to other types of pathogens, such as viruses and intracellular bacteria, respectively. Option D is related to the immune evasion strategy of encapsulated bacteria, such as Streptococcus pneumoniae, which is a common cause of community-acquired pneumonia. However, the patient's response to cephalexin does not suggest that the causative pathogen is an encapsulated bacterium. </think><answer>C. Peptidoglycan is its major antigen that induces an immune response. </answer>